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Exercise sheet 1.1

Understanding Neuroscience Papers I

This exercise builds on the paper "Experience-dependent spatial expectations in mouse visual cortex" (Fiser et al., 2016 Nature Neuroscience).

Make sure to upload your answers in PDF format to the Moodle by
11:59 pm on Tuesday 22. September 2026.
Please provide your name, student ID, and email address in the header. Use font size 12 and a line spacing of 1.5 with justified paragraphs and a maximum of 2 pages. Submit your answers individually. Submissions that are late or have copied text from the paper or peers **will not** be considered.

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1. In Fiser et al., how does the classification performance of grating type (A or B) and location change from condition 1 to 4 in V1 and CA1? (Choose the 1 correct answer)

   ☐ a) In both V1 and CA1, performance does not change from condition 1 to 4.

   ☐ b) In both V1 and CA1, performance increases.

   ☐ c) In both V1 and CA1, performance decreases.

   ☒ d) In V1, performance increases. In CA1, performance decreases.

   ☐ e) In V1, performance decreases. In CA1, performance increases.

2. Which of these statements is true about V1 predictive and visual neurons? (Choose the 1 correct answer)

   ☐ a) Individual predictive neurons are equally tuned to A and B gratings.

   ☐ b) Individual visual neurons are equally tuned to A and B gratings.

   ☐ c) Predictive neurons are only active after trials when subjects encountered an unexpected grating.

   ☐ d) Visual neurons are only active after trials when subjects encountered an unexpected grating.

   ☒ e) The activity of predictive neurons increases before grating onset, whereas the activity of visual neurons increases during the grating.

3. **What** did you find most interesting about this paper and **Why**? (1-2 paragraphs; at least 4 sentences total)

Honestly, the omission experiment (Fig. 4) is what got me, the fact that just not showing the grating still gets you a strong, selective V1 response, and that this response scales with how strong the predictive activity was beforehand. It felt quiet strong evidence, way more convincing to me than the classification accuracy numbers, because it's the one result where there's literally no stimulus for the brain to be responding to. It has to be internally generated. I also liked that they didn't stop there, they checked the omission selective neurons against expected gray screen at the start/end of the tunnel and got nothing, which rules out the boring explanation that this is just some generic "stimulus offset/onset" artifact.

I also kinda like the check of the "running speed control" described in the Methods (where they trained the location classifier on slow transversal and tested it on fast ones, and vice versa). It's a small, kinda unglamorous check compared to the flashy accuracy numbers in Fig. 1e, but it was the kind of thing that decided whether I actually believed the "V1 becomes spatially informative with experience" headline or just believed the classifier learned something correlated with location. I've been wrestling with my own classifier this past semesters to have a lot of appreciation for people who bother running that kind of boring but load bearing control instead of just reporting the number that looks good.